



AI-DRIVEN PREDICTION OF HEAT TRANSFER COEFFICIENTS FOR GAS TURBINE BLADE COOLING

Semester Project

École Polytechnique Fédérale de Lausanne
School of Engineering
MSc Mechanical Engineering
Neuchâtel, June 2026

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ABSTRACT

This report presents the development and evaluation of two machine learning surrogate models for the prediction of heat transfer and flow fields in rib-turbulated internal cooling channels of gas turbine blades. The first model is a Physics-Informed Neural Network (PINN), trained simultaneously on 50 Computational Fluid Dynamics (CFD) design points to predict all flow fields: velocity in x , y and z , pressure, and temperature. The second model is a purely data-driven Multilayer Perceptron (MLP), trained to predict the local wall heat flux from the same inputs. Together, the two models form a surrogate pipeline capable of predicting the Heat Transfer Coefficient (HTC) at any point in the five-dimensional design space, enabling rapid evaluation without recourse to full CFD simulations.

The design space is parametrized by the channel aspect ratio (AR), rib angle (α), rib-to-hydraulic diameter ratio (e/D_h), rib pitch-to-height ratio (P/e), and Reynolds number (Re), with 50 cases sampled via Latin Hypercube Sampling (LHS) to ensure good coverage. The PINN is formulated using the steady-state incompressible Navier–Stokes equations coupled with the energy equation, all posed in dimensionless form.

Results on four unseen design points show that the PINN recovers the temperature field with approximately 11% mean relative error, the velocity field with approximately 25% error, and the pressure with approximately 8% error. The MLP predicts the heat flux distribution with 13–32% error depending on the design point. The combined HTC prediction reaches 30–51% error, with large-scale inter-design-point variation correctly preserved. Both models complete inference in under 20 seconds on CPU, compared to hours for a full CFD simulation.

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1 INTRODUCTION

1.1 ENGINEERING CONTEXT: INTERNAL COOLING OF GAS TURBINE BLADES

Gas turbine blades operate in some of the most thermally demanding environments in mechanical engineering. The hot combustion gases flowing over the external blade surface can exceed 1400°C , well above the melting point of current nickel superalloys. One of the most effective approaches to prevent blade failure is the use of **rib-turbulated internal cooling channels**, in which ribs (also called turbulators) are into the internal walls of the blade. These ribs disrupt boundary layer development, promote secondary flows and recirculation zones, and thereby substantially enhance convective heat transfer between the cooling fluid and the blade wall.

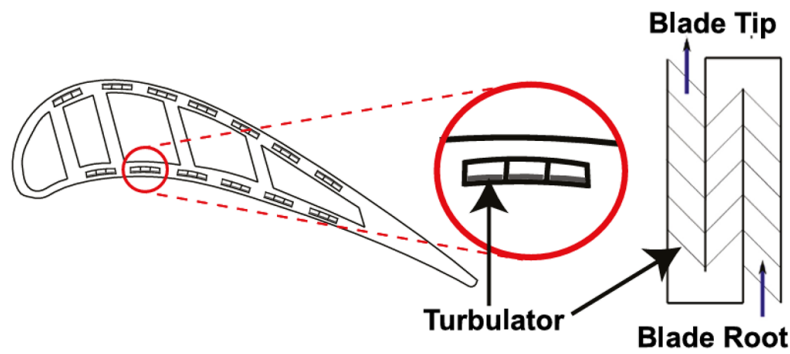


Fig. 1. Conceptual illustration of wall-integrated cooling passages in a turbine blade, reproduced from [3]. Left: top view of a blade showing cooling passages embedded within the blade wall. Center: enlarged top view of an individual channel highlighting the rib turbulators. Right: schematic side view of the ribbed cooling channel along the blade span (root to tip). The drawings are schematic and not to scale.

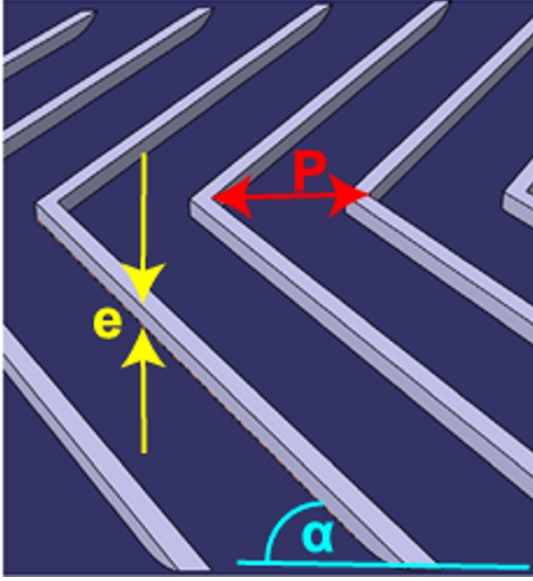
Figure 1: Conceptual illustration of wall-integrated cooling passages in a gas turbine blade. Left: top view of a blade showing embedded cooling passages. Centre: enlarged view highlighting the rib turbulators. Right: schematic side view of the ribbed cooling channel from root to tip [12].

The fundamental trade-off is unavoidable: while ribs enhance heat transfer, they also increase the pressure drop across the cooling channel, raising the pumping power required to drive the coolant. Effective blade cooling design therefore requires the simultaneous optimisation of heat transfer enhancement and pressure drop penalty, a fundamentally multi-objective problem [8].

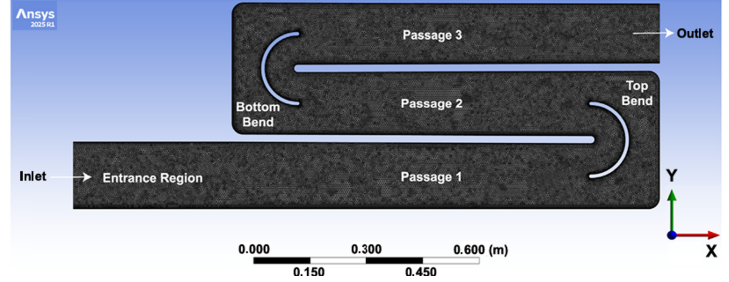
1.2 DESIGN SPACE AND CFD DATASET

The cooling channel geometry and flow conditions are parametrised by five independent variables:

1. **Aspect ratio** (AR): channel width to height ratio, ranging from 4 to 15.
2. **Rib angle** (α): angle of the V-rib with respect to the streamwise direction, ranging from 30° to 75° .
3. **Rib height ratio** (e/D_h): rib height to hydraulic diameter, 5% to 20%.
4. **Rib pitch ratio** (P/e): centre-to-centre rib spacing to rib height, ranging from 5 to 15.
5. **Reynolds number** (Re): ranging from 20,000 to 200,000.



(a) V-rib geometry parameters: height e , pitch P , and angle α .



(b) Three-pass G180° guide-vane CFD geometry meshed in ANSYS 2025 R1, showing the inlet, three straight passages, two 180° bends, and the outlet.

Figure 2: Design space parametrisation and CFD geometry [12]

A dataset of 50 CFD cases was generated by sampling this five-dimensional design space using Latin Hypercube Sampling (LHS) [16, 1], a stratified strategy ensuring uniform one-dimensional projections and good coverage of the parameter space. Each simulation was run on a three-pass G180° guide-vane configuration in ANSYS Fluent, producing meshes of ~ 10 million cells with all five output fields (v_x , v_y , v_z , p , T).

1.3 PROJECT OBJECTIVES AND GENERAL STRATEGY

The central objective is to replace expensive CFD simulations ($\sim$ hours each) with fast surrogate models ($\sim$ seconds) for integration into a multi-objective optimisation loop. Two competing objectives are targeted:

- **Minimise pressure drop:** $\Delta P = P_{\text{in}} - P_{\text{out}}$
- **Maximise heat transfer coefficient:** $\text{HTC} = q'' / (T_w - T_b)$

Two complementary models are developed: a **Physics-Informed Neural Network (PINN)** [20] predicting all five flow fields across the 3D domain (from which ΔP and T_b are extracted), and a **data-driven MLP** predicting the local wall heat flux q'' directly from CFD data.

2 LITERATURE REVIEW

2.1 PHYSICS-INFORMED NEURAL NETWORKS

Physics-Informed Neural Networks (PINNs), introduced by Raissi, Perdikaris, and Karniadakis [20], embed the governing PDEs of a physical system directly into the training loss of a neural network via automatic differentiation. This allows the network to be trained with little or no labelled data while remaining constrained to physically consistent solutions, covering both forward problems (predicting fields from boundary conditions) and inverse problems (inferring parameters from sparse measurements).

The approach was quickly extended to incompressible flow. Jin et al. [13] demonstrated that PINNs can solve steady and unsteady Navier–Stokes equations at moderate Reynolds numbers. Eivazi et al. [6] extended the framework to Reynolds-Averaged Navier–Stokes (RANS) equations, covering the turbulent regime relevant to GT cooling ($Re \sim 10^4$ – 10^5). Cai et al. [4] established PINNs as a tool for heat transfer problems, recovering temperature fields and heat flux from partial data in convective settings. A comprehensive review of PINN developments in complex fluid dynamics is given in [25].

2.2 PINNs WITH SPARSE DATA AND GEOMETRY-AWARENESS

Hanrahan et al. [9] showed that PINNs trained with sparse observations substantially improve field reconstruction over purely data-driven interpolation, which motivates the supervised component used here. For 3D thermal problems, Bhatnagar et al. [3] showed that PINNs trained on sparse domain data can recover velocity and temperature fields in complex geometries.

Providing geometric information to PINNs over variable geometries is an open challenge. Oldenburg et al. [17] addressed this with the Geometry-Aware PINN (GAPINN), encoding geometric information through an additional network branch. The present work follows a related but simpler philosophy: sigmoid-based wall-position features are used to inform the network of the multi-pass channel structure, providing geometric awareness without requiring a separate encoder.

2.3 PINNs FOR GAS TURBINE APPLICATIONS

Mai et al. [15] applied PINNs to reconstruct 2D temperature fields in turbine blades from sparse thermocouple data, directly relevant to the inverse problem motivating the GT cooling setting. Puttock-Brown et al. [19] applied PINNs to the inverse heat transfer problem in rotating cavities, showing they can identify unknown HTC from accessible temperature data.

2.4 DATA-DRIVEN SURROGATES FOR INTERNAL COOLING

Parallel to PINNs, purely data-driven surrogate models have been extensively applied to internal cooling optimisation. Wang et al. [23] trained supervised models on CFD simulations of GT blade cooling channels for fast prediction of thermal and aerodynamic performance metrics. Kwon et al. [14] combined response surface methods with neural networks to optimise rib geometry. Sahin et al. [21] applied deep operator learning to build surrogate models with uncertainty quantification for internal cooling channel rib profiles. Xi et al. [24] and Seo et al. [22] demonstrated neural networks and genetic algorithms for multi-objective optimisation of ribbed channel geometry.

2.5 LATIN HYPERCUBE SAMPLING

The 50-point CFD dataset was generated using Latin Hypercube Sampling (LHS) [16]. Afzal et al. [1] confirmed that LHS-based designs lead to better surrogate accuracy than factorial designs at the same sample budget in turbomachinery applications.

3 PHYSICS-INFORMED NEURAL NETWORK

3.1 PINN FORMULATION

A Physics-Informed Neural Network (PINN) [20] trains a neural network to satisfy both the governing PDEs of a physical system and available labelled data simultaneously. PDE residuals are evaluated via automatic differentiation through the network and included as loss terms during training, constraining the solution to be physically consistent across the full domain. The motivation for using a PINN here is **generalisation**: the goal is to train a single model on a limited set of 50 design points that can predict accurate flow fields for unseen geometries. Rather than fitting the full CFD dataset of each design point, only a sparse subset of 500 supervised points per design point is provided, forcing the model to rely on the physics to fill in the rest of the domain. This trade-off is intentional: overfitting the available CFD data would reduce the model's ability to generalise to new design parameters.

The problem is **steady-state**, so no initial condition loss is required. The training loss combines three families of terms: PDE residuals at collocation points Ω_c distributed across the domain interior, boundary condition terms enforced at the domain surfaces Γ , and supervised data terms at the sparse set Ω_{sup} . Spatial derivatives are computed by automatic differentiation: first-order gradients by a first backward pass, and second-order Laplacian terms by differentiating each component of the first-order gradient again along its corresponding spatial dimension.

3.2 NON-DIMENSIONALISATION

All physical fields are non-dimensionalised before training. The reference scales are defined per design point:

$$\tilde{v}_i = v_i / V_{\text{in}}, \quad i = x, y, z \quad (3.1)$$

$$\tilde{p} = (p - p_{\text{ref}}) / \sigma_p, \quad p_{\text{ref}} = \bar{p}_{\text{outlet}}^{\text{CFD}}, \quad \sigma_p = \text{std}(p_{\text{CFD}}) \quad (3.2)$$

$$\tilde{T} = (T - T_w) / (T_{\text{in}} - T_w) \quad (3.3)$$

The choice of σ_p (the standard deviation of the full CFD pressure field) rather than the dynamic pressure ρV_{in}^2 is important. At high rib blockage ratios or high friction Reynolds numbers, the viscous pressure drop can substantially exceed the dynamic pressure, causing $\tilde{p}$ to have a range of order 10^2 – 10^3 if ρV_{in}^2 is used as the reference. This would cause the supervised pressure loss to dominate all other loss terms regardless of their physical importance. Using σ_p ensures $\tilde{p} \sim \mathcal{O}(1)$ for every design point by construction.

The temperature normalisation gives $\tilde{T} = 1$ at the inlet and $\tilde{T} = 0$ at the wall. In the experimental setup, warm air is injected at $T_{\text{in}} \approx 329$ K and the channel walls are maintained cold at $T_w \approx 293$ K, so the fluid is progressively cooled as it flows through the channel. The wall temperature is identified as $T_w = \min(T_{\text{CFD}})$.

This normalisation is only meaningful when there is an actual temperature difference between the fluid and the walls. The PINN was first developed and tested on an isothermal channel (Section 3.8.1), where $T_{\text{in}} = T_w$. In that setting, the temperature field is trivially uniform and the denominator in Eq. (3.3) collapses to zero, making the normalisation undefined and the loss numerically unstable. This observation motivated the decision to restrict the non-isothermal PINN to design points with $|T_{\text{in}} - T_w| \geq 5$ K and to treat the isothermal case separately.

3.3 GOVERNING EQUATIONS

The flow is modelled as steady, incompressible, and Newtonian, with temperature-dependent viscosity following Sutherland's law. The governing equations in dimensionless form are:

$$\nabla \cdot \tilde{\mathbf{u}} = 0 \quad (3.4)$$

$$\tilde{\mathbf{u}} \cdot \nabla \tilde{u}_i + \partial_{x_i} \tilde{p} - \frac{\nu \tilde{\mu}(\tilde{T})}{V_{\text{in}}} \nabla^2 \tilde{u}_i = 0, \quad i = x, y, z \quad (3.5)$$

$$\tilde{\mathbf{u}} \cdot \nabla \tilde{T} - \frac{\alpha_{\text{th}}}{V_{\text{in}}} \nabla^2 \tilde{T} = 0 \quad (3.6)$$

where $\tilde{\mu} = \mu(\tilde{T})/\mu_{\text{ref}}$ is the dimensionless dynamic viscosity from Sutherland's law:

$$\mu(T) = \mu_{\text{ref}} \left(\frac{T}{T_{\text{ref}}} \right)^{3/2} \frac{T_{\text{ref}} + S}{T + S}, \quad \mu_{\text{ref}} = 1.716 \times 10^{-5} \text{ Pa}\cdot\text{s}, \quad T_{\text{ref}} = 278.15 \text{ K}, \quad S = 110.4 \text{ K} \quad (3.7)$$

The dimensionless viscous coefficient ν/V_{in} and thermal diffusivity coefficient $\alpha_{\text{th}}/V_{\text{in}}$ carry units of metres, scaling the diffusion terms relative to the domain size. Both are computed per design point from the inlet conditions and fluid properties.

A note on the non-dimensionalisation of the NS equations: substituting $v_i = V_{\text{in}} \tilde{v}_i$ and $p = p_{\text{ref}} + \sigma_p \tilde{p}$ into the dimensional momentum equation gives:

$$V_{\text{in}}^2 (\tilde{\mathbf{u}} \cdot \nabla \tilde{u}_i) = -\frac{\sigma_p}{\rho} \partial_{x_i} \tilde{p} + \nu \tilde{\mu} V_{\text{in}} \nabla^2 \tilde{u}_i \quad (3.8)$$

Dividing through by V_{in}^2 yields the pressure gradient coefficient $\sigma_p/(\rho V_{\text{in}}^2)$. When $\sigma_p = \rho V_{\text{in}}^2$, this reduces to unity and the equation is in standard form. Here, $\sigma_p = \text{std}(p_{\text{CFD}})$ is used instead, so this coefficient is not necessarily unity but remains $\mathcal{O}(1)$ by construction of σ_p .

3.4 LOSS FUNCTION

The total PINN loss combines three families of terms:

$$\begin{aligned} \mathcal{L} = & w_{\text{pde}} \underbrace{(\mathcal{L}_{\text{div}} + \sum_i \mathcal{L}_{\text{mom},i} + \mathcal{L}_{\text{heat}})}_{\mathcal{L}_{\text{pde}}} \\ & + w_{\text{bc}} \underbrace{(\sum_i \mathcal{L}_{\text{in},u_i} + \mathcal{L}_{\text{in},T} + \mathcal{L}_{\text{out},p} + \sum_i \mathcal{L}_{\text{wall},u_i} + \mathcal{L}_{\text{wall},T})}_{\mathcal{L}_{\text{bc}}} \\ & + w_{\text{sup}} \underbrace{(\sum_i \mathcal{L}_{\text{sup},u_i} + \mathcal{L}_{\text{sup},p} + \mathcal{L}_{\text{sup},T})}_{\mathcal{L}_{\text{sup}}} \end{aligned} \quad (3.9)$$

All supervised and BC losses are normalised by the field variance σ_ϕ^2 computed from CFD training data, ensuring all terms are $\mathcal{O}(1)$. The PDE losses are scaled by global normalisation factors derived from the coordinate and output standard deviations computed over all training design points. The inlet velocity profile BC is enforced using the actual CFD inlet velocity profile extracted from each design point. The full loss term definitions are given in Table 1.

Table 1: PINN loss terms. $\overline{(\cdot)}_\Omega$ denotes the mean squared value over points in domain Ω . σ_ϕ^2 is the variance of field $\tilde{\phi}$ from CFD training data. $\tilde{\mu}(\tilde{T})$ is the Sutherland viscosity ratio evaluated from the current predicted temperature.

Category	Name	Notation	Formula
PDE	Continuity	$\mathcal{L}_{\text{div}}$	$\overline{(\nabla \cdot \tilde{\mathbf{u}})^2}_{\Omega_c}$
	Navier-Stokes ($i=x, y, z$)	$\mathcal{L}_{\text{mom},i}$	$\overline{\left(\tilde{\mathbf{u}} \cdot \nabla \tilde{u}_i + \frac{\sigma_p}{\rho V_{\text{in}}^2} \partial_{x_i} \tilde{p} - \frac{\nu \tilde{\mu}}{V_{\text{in}}} \nabla^2 \tilde{u}_i \right)^2}_{\Omega_c}$
	Energy	$\mathcal{L}_{\text{heat}}$	$\overline{\left(\tilde{\mathbf{u}} \cdot \nabla \tilde{T} - \frac{\alpha_{\text{th}}}{V_{\text{in}}} \nabla^2 \tilde{T} \right)^2}_{\Omega_c}$
BC	Inlet velocity	$\mathcal{L}_{\text{in},u_i}$	$\frac{1}{\sigma_{\tilde{u}_i}^2} \overline{(\tilde{u}_i - \tilde{u}_i^{\text{CFD}})^2}_{\Gamma_{\text{in}}}$
	Inlet temperature	$\mathcal{L}_{\text{in},T}$	$\frac{1}{\sigma_{\tilde{T}}^2} \overline{(\tilde{T} - 1)^2}_{\Gamma_{\text{in}}}$
	Outlet pressure	$\mathcal{L}_{\text{out},p}$	$\frac{1}{\sigma_{\tilde{p}}^2} \overline{\tilde{p}^2}_{\Gamma_{\text{out}}}$
	Wall no-slip	$\mathcal{L}_{\text{wall},u_i}$	$\frac{1}{\sigma_{\tilde{u}_i}^2} \overline{\tilde{u}_i^2}_{\Gamma_w}$
	Wall temperature	$\mathcal{L}_{\text{wall},T}$	$\frac{1}{\sigma_{\tilde{T}}^2} \overline{\tilde{T}^2}_{\Gamma_w}$
Supervised	Velocity	$\mathcal{L}_{\text{sup},u_i}$	$\frac{1}{\sigma_{\tilde{u}_i}^2} \overline{(\tilde{u}_i - \tilde{u}_i^{\text{CFD}})^2}_{\Omega_{\text{sup}}}$
	Pressure	$\mathcal{L}_{\text{sup},p}$	$\frac{1}{\sigma_{\tilde{p}}^2} \overline{(\tilde{p} - \tilde{p}^{\text{CFD}})^2}_{\Omega_{\text{sup}}}$
	Temperature	$\mathcal{L}_{\text{sup},T}$	$\frac{1}{\sigma_{\tilde{T}}^2} \overline{(\tilde{T} - \tilde{T}^{\text{CFD}})^2}_{\Omega_{\text{sup}}}$

3.5 LOSS WEIGHTING STRATEGY

Three strategies were evaluated to balance the competing loss terms:

- **Manual weighting.** Fixed scalar weights w_{pde} , w_{bc} , w_{sup} are hand-tuned by trial and error. This approach is tedious and case-specific: weights calibrated for one geometry or Reynolds number do not transfer to another design point.
- **Learnable weights.** The loss weights are treated as jointly-optimised parameters with min/max bounds, following a Lagrangian-style formulation:

$$\mathcal{L}_{\text{tot}} = w_{\text{pde}} \mathcal{L}_{\text{pde}} + w_{\text{bc}} \mathcal{L}_{\text{bc}} + w_{\text{sup}} \mathcal{L}_{\text{sup}} - w_{\text{pde}} - w_{\text{bc}} - w_{\text{sup}} \quad (3.10)$$

where the network weights are minimised over θ while the loss weights are maximised. In practice, the weights collapsed toward their bounds, providing no meaningful gain over fixed weights.

- **SoftAdapt** [10]. Dynamic rescaling of weights by the rate of change of each loss component across iterations. This produced marginal improvement but added hyperparameter sensitivity.

The conclusion is that adaptive weighting is, at this stage, unnecessary. Proper non-dimensionalisation and variance-normalisation of all loss terms naturally produces $\mathcal{O}(1)$ values for all components across all 50 design points with uniform weights $w = 1$.

3.6 NETWORK ARCHITECTURE

The network is a fully-connected feedforward network with **Sine activations** throughout all hidden layers and a linear output layer. Sine activations provide the smooth, twice-differentiable function class required for computing second-order Laplacian terms via automatic differentiation. Xavier normal initialisation is applied to all linear layers.

The input concatenates the physical coordinates with the five design parameters, z-score normalised over all 50 training design points:

$$\mathbf{x}_{\text{raw}} = (x, y, z, AR_{\text{nd}}, (e/D_h)_{\text{nd}}, (P/e)_{\text{nd}}, \alpha_{\text{nd}}, Re_{\text{nd}}) \in \mathbb{R}^8 \quad (3.11)$$

Table 2: Design parameter z-score normalisation statistics.

Parameter	AR	e/D_h	P/e	α ($^\circ$)	Re
Mean	9.0	0.127	10.0	52.0	108,000
Std	3.5	0.048	3.1	14.5	58,000

The sigmoid wall-encoding features s_1 and s_2 (Section 3.8.3) are appended to give a 10-dimensional raw input, which is then z-score normalised globally using statistics derived from the full CFD training dataset before being passed to the network.

The output is the five-component dimensionless field vector $\hat{\mathbf{u}} = (\tilde{v}_x, \tilde{v}_y, \tilde{v}_z, \tilde{p}, \tilde{T}) \in \mathbb{R}^5$. When CFD data is available for a specific design point, the output can be converted back to physical units using the per-DP reference scales (Eqs. (3.1)–(3.3)). For design points with no CFD data at all, the model directly returns the normalised fields $(\tilde{v}_i, \tilde{p}, \tilde{T})$, from which physical quantities can be reconstructed given only the inlet conditions $V_{\text{in}}, T_{\text{in}}, T_w$, and σ_p .

3.7 OPTIMISER AND TRAINING SCHEDULE

Several optimisers were explored. L-BFGS was tested as a second-order method expected to provide fast convergence to high-precision solutions. A two-stage strategy was also attempted: Adam for global exploration in early training, followed by L-BFGS for fine-tuning. Neither approach improved generalisation to unseen design points meaningfully over Adam alone, while substantially increasing implementation complexity. The final training uses the **Adam optimiser** with $\beta_1 = 0.99$, $\beta_2 = 0.999$ and an exponential learning rate decay:

$$lr_e = lr_0 \cdot \gamma^e, \quad \gamma = \left(\frac{lr_{\text{end}}}{lr_0} \right)^{1/E} \quad (3.12)$$

where E is the total number of epochs. The training loop processes all 50 design points in randomly shuffled order each epoch, accumulating gradients from all forward-backward passes before a single optimiser step. This ensures each parameter update incorporates information from the full set of design points, reducing gradient noise.

A hyperparameter search was conducted on the EPFL IZAR HPC cluster using Weights & Biases Bayesian sweeps (details in Appendix A). Training with the selected configuration takes 2–10 hours

on the cluster depending on the number of epochs and collocation points; inference over a new design point takes 1–20 seconds depending on model size.

The selected hyperparameters are: 5,000 epochs, hidden dimension 128, 6 layers ($\approx 240,000$ parameters), $lr_0 = 3.6 \times 10^{-3}$, decay ratio 0.09, $n_{\text{train}} = 10,000$ collocation points per epoch per design point (10 hours of training, 20 seconds for inference).

3.8 PROGRESSIVE MODEL DEVELOPMENT: V1, V2, V3

3.8.1 PINN V1: SIMPLE ISOTHERMAL CHANNEL

The first version was developed and validated on a straight, isothermal, single-passage channel as a proof of concept. The network takes only (x, y, z) as input. Two conditions were compared: training with 500 supervised CFD points and physics-only training (no supervised data).

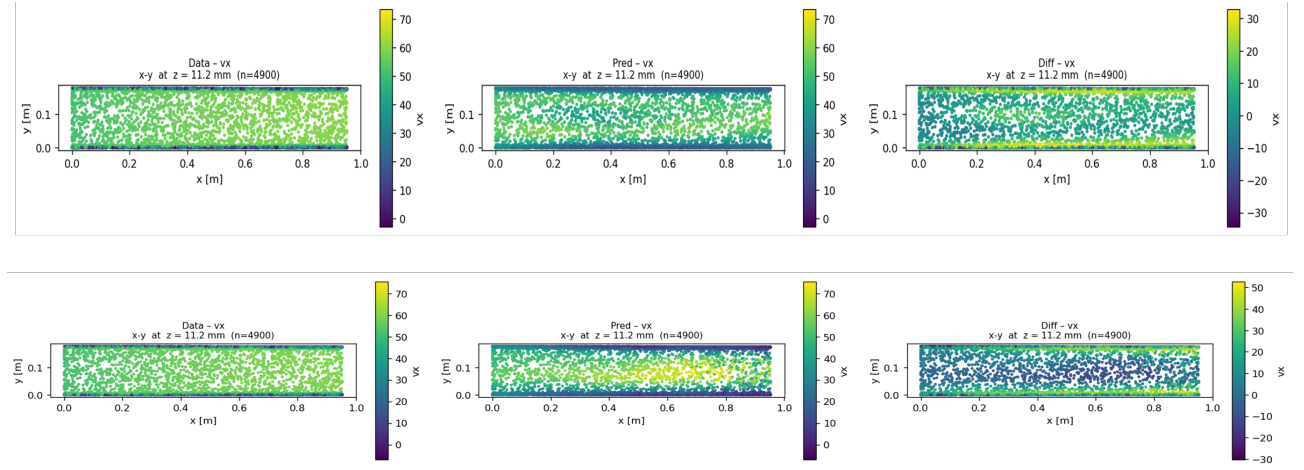


Figure 3: PINN V1: velocity field v_x (x-y cross-section at $z = 11.2$ mm). Top: with 500 supervised CFD points. Bottom: physics-only. Columns: CFD reference, PINN prediction, pointwise difference. Near-wall errors reflect the CFD cell-centre offset from the wall face.

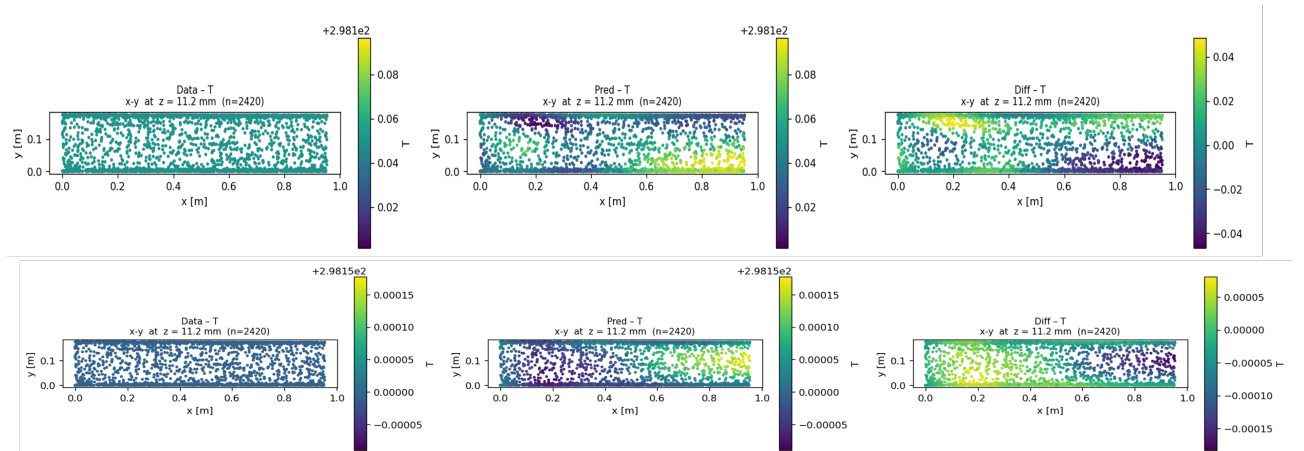


Figure 4: PINN V1: temperature field. The isothermal channel has uniform temperature and the physics losses alone recover it exactly. Absolute errors are well below 0.01°C in both cases, and are in fact smaller without supervised data because the trivial solution is already a global minimum.

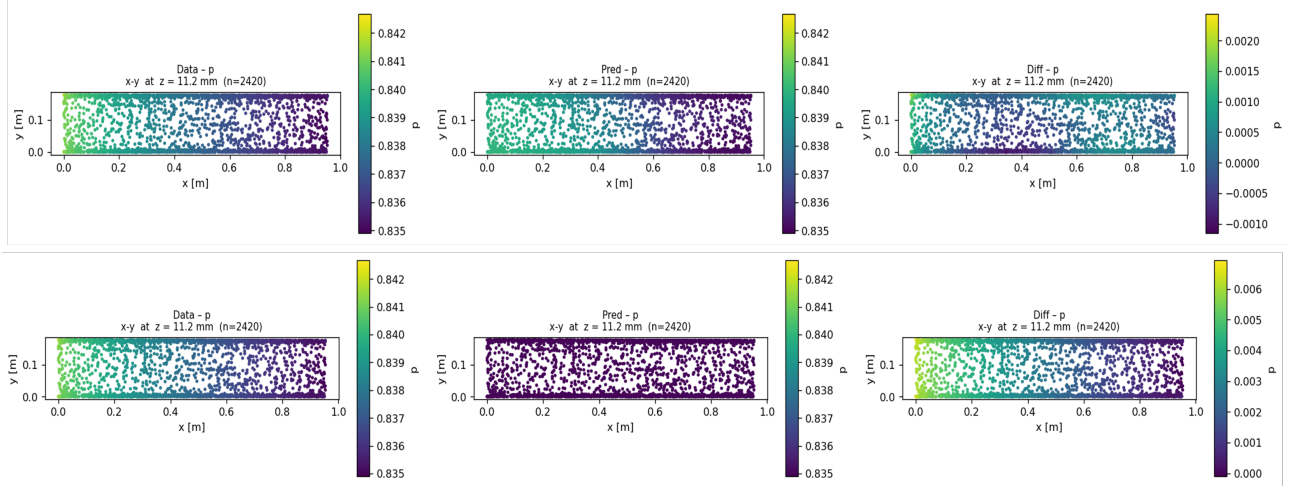


Figure 5: PINN V1: pressure field. With 500 supervised points (top): errors ≈ 0.001 Pa. Without supervised data (bottom): the model converges to a constant pressure field. A constant pressure field satisfies the NS momentum equations (the gradient is zero, so the advection-diffusion balance is not violated) and simultaneously minimises the outlet BC loss (since $\bar{p}^2_{\Gamma_{\text{out}}} = 0$ for $\bar{p} \equiv 0$). Sparse supervised data is therefore essential to select the physically correct non-uniform pressure solution.

3.8.2 PINN V2: THREE-PASSAGE GUIDE-VANE WITH PARAMETRIC INPUTS

Version 2 extended the model to the full three-passage guide-vane geometry and introduced parametric conditioning by appending the five normalised design parameters to the spatial input. A single PINN trained on all 50 design points simultaneously serves as a parametric surrogate over the entire design space.

Adaptive near-wall collocation sampling. Naive uniform volume sampling produces poor boundary condition enforcement because the near-wall regions, which carry most of the physical information about no-slip and wall temperature, represent a tiny fraction of the domain volume. An adaptive strategy was implemented, splitting the volume pool into three sub-pools drawn from the STL mesh:

1. **Uniform interior points** (50% of pool): sampled uniformly from the mesh volume.
2. **Near-horizontal-wall points** (15%): surface points on horizontal wall faces ($|n_z| \geq 0.7$) displaced inward by a random distance $\varepsilon \sim \mathcal{U}(0, \delta_{\text{horiz}})$ along the inward face normal, $\delta_{\text{horiz}} = 1$ mm.
3. **Near-vertical-wall points** (35%): same procedure on vertical faces ($|n_z| < 0.7$), with $\delta_{\text{vert}} = 20$ mm.

The thresholds reflect the channel dimensions: $\delta_{\text{horiz}}/H \approx 0.8\%$ and $\delta_{\text{vert}}/L \approx 1.3\%$, keeping the near-wall fractions comparable relative to each direction. Both pools are pre-computed once from the STL mesh and cached to disk; at each epoch, n_{train} points are sampled without replacement.

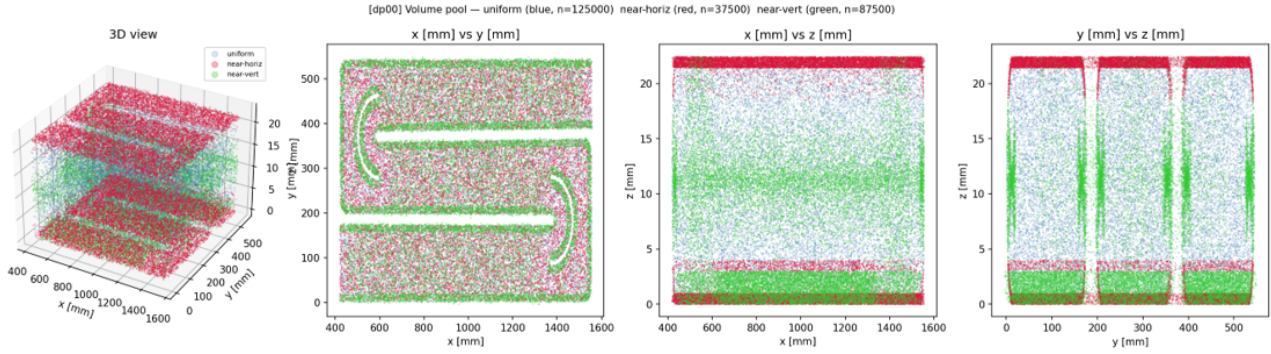


Figure 6: Adaptive collocation point distribution for design point dp00. Blue: uniform interior points ($n = 125,000$). Red: near-horizontal-wall points ($n = 37,500$). Green: near-vertical-wall points ($n = 87,500$). The three projections confirm dense near-wall coverage while maintaining interior coverage.

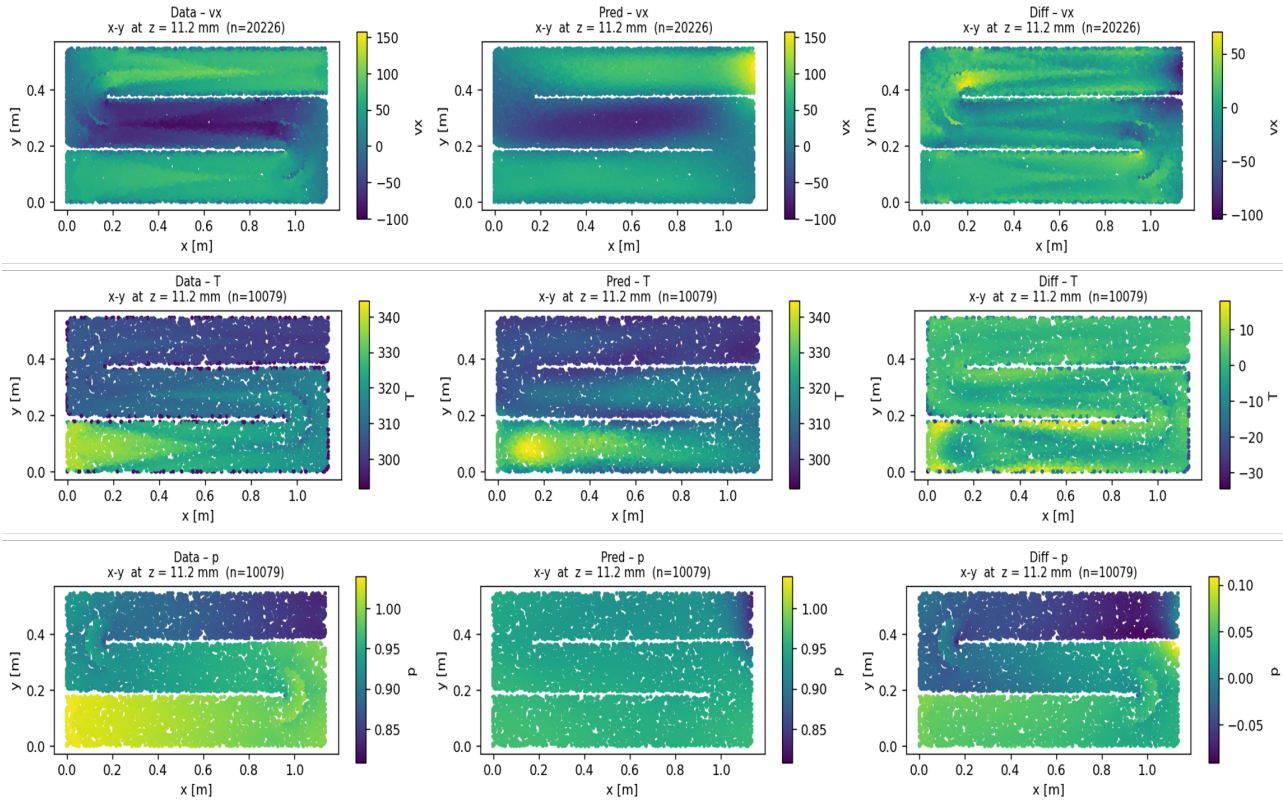


Figure 7: PINN V2 fields on one unseen design point (x-y slice at $z = 11.2$ mm). Top: v_x , well recovered. Middle: T , well recovered. Bottom: p , incorrect. Predicted pressure contours pass through the internal walls, ignoring the guide-vane geometry entirely. Velocity and temperature are strongly constrained by the NS equations, the inlet profile BC, and the wall BCs; pressure is only constrained by the single outlet pressure BC and its gradient coupling in the momentum equations.

V2 pressure failure. With five design parameters added as inputs, velocity and temperature were well recovered on unseen design points. However, the pressure field failed completely: the predicted field passed through the solid internal walls with no discontinuity, as if the channel had no guide vanes. The root cause is the imbalance in how strongly each output field is constrained by the loss. Velocity is anchored by three NS equations, three inlet velocity BCs, and three wall no-slip conditions.

Temperature is anchored by the energy equation, the inlet temperature BC, and the wall temperature BC. Pressure, in contrast, has only the single outlet pressure BC and the pressure gradient term within the momentum equations. Without any geometric signal distinguishing the three passages, the network assigns a single smooth pressure field across the whole domain.

3.8.3 PINN V3: SIGMOID WALL-POSITION ENCODING

To give the network explicit passage-membership information, two sigmoid features are appended to the input:

$$s_1 = \sigma\left(\frac{y - y_{\text{wall}_1}}{\varepsilon}\right), \quad s_2 = \sigma\left(\frac{y - y_{\text{wall}_2}}{\varepsilon}\right), \quad \varepsilon = 0.002 \text{ m} \quad (3.13)$$

with internal wall positions $y_{\text{wall}_1} = 0.185 \text{ m}$ and $y_{\text{wall}_2} = 0.375 \text{ m}$, fixed and shared across all 50 design points since the channel height is the same for all. The width $\varepsilon = 0.002 \text{ m}$ was chosen so that the sigmoid transitions sharply enough to clearly separate the three passages while remaining differentiable for the autograd computation. The value $\varepsilon = 0.002 \text{ m}$ corresponds to the sigmoid rising from 0.05 to 0.95 over a distance of roughly $2 \times \varepsilon \ln(19) \approx 12 \text{ mm}$, which is small relative to the channel height (120 mm) and on the order of a rib height.

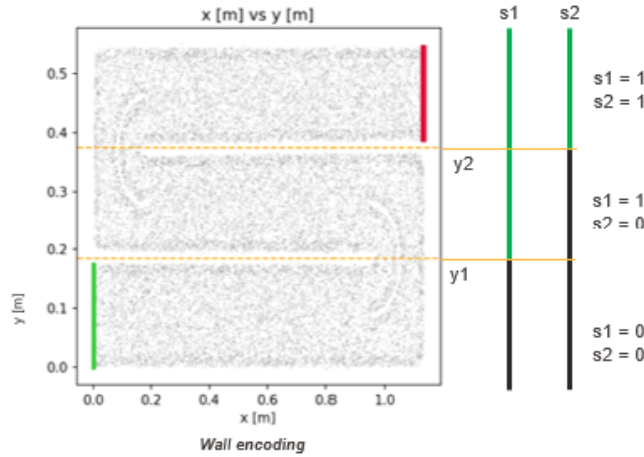


Figure 8: Sigmoid wall encoding. Orange dashed lines mark the two internal wall positions y_1 and y_2 . The feature pair (s_1, s_2) takes values $(0, 0)$ in Passage 1, $(1, 0)$ in Passage 2, and $(1, 1)$ in Passage 3, giving the network unambiguous passage membership at every input point.

With this encoding, a point in Passage 1 and a point in Passage 2 at the same (x, z) coordinate receive different (s_1, s_2) values, allowing the network to represent distinct pressure levels in each passage. The full V3 input vector is:

$$\mathbf{x}_{\text{in}} = (x, y, z, AR_{\text{nd}}, (e/D_h)_{\text{nd}}, (P/e)_{\text{nd}}, \alpha_{\text{nd}}, Re_{\text{nd}}, s_1, s_2) \in \mathbb{R}^{10} \quad (3.14)$$

which is z-score normalised globally before being passed to the network.

3.9 ERROR METRIC

Results are reported using the mean relative error $\bar{\varepsilon}$, defined for a scalar field ϕ as:

$$\bar{\varepsilon} = \frac{1}{N} \sum_{i=1}^N \frac{|\phi_{\text{pred}}^{(i)} - \phi_{\text{CFD}}^{(i)}|}{|\phi_{\text{CFD}}|} \times 100 \text{ (\%)} \quad (3.15)$$

where the denominator is the mean absolute value of the CFD field over all evaluation points, avoiding division by zero at nodes where the local CFD value is small.

3.10 RESULTS ON UNSEEN DESIGN POINTS

PINN V3 is evaluated on 4 design points held out from training (dp102, dp103, dp104, dp105). All errors are computed using the mean relative error $\bar{\varepsilon}$ of Eq. (3.15) over all volume points of each test design point.

3.10.1 TEMPERATURE FIELD

The normalised temperature field $\theta = (T - T_w)/(T_{in} - T_w)$ is recovered with mean relative errors of 12.7%, 12.0%, 10.9%, and 10.4% for dp102–dp105 respectively. The global thermal stratification is well captured in all cases: the fluid enters warm at $\theta = 1$ and cools progressively toward the walls as it travels through the three passages. The streamwise temperature gradient within each passage is correctly reproduced, and the model correctly identifies which passage is hotter at any given axial location. Errors concentrate near the bend regions, where the sharp flow reversal and associated secondary flows create steep local temperature gradients that the smooth network cannot fully resolve.

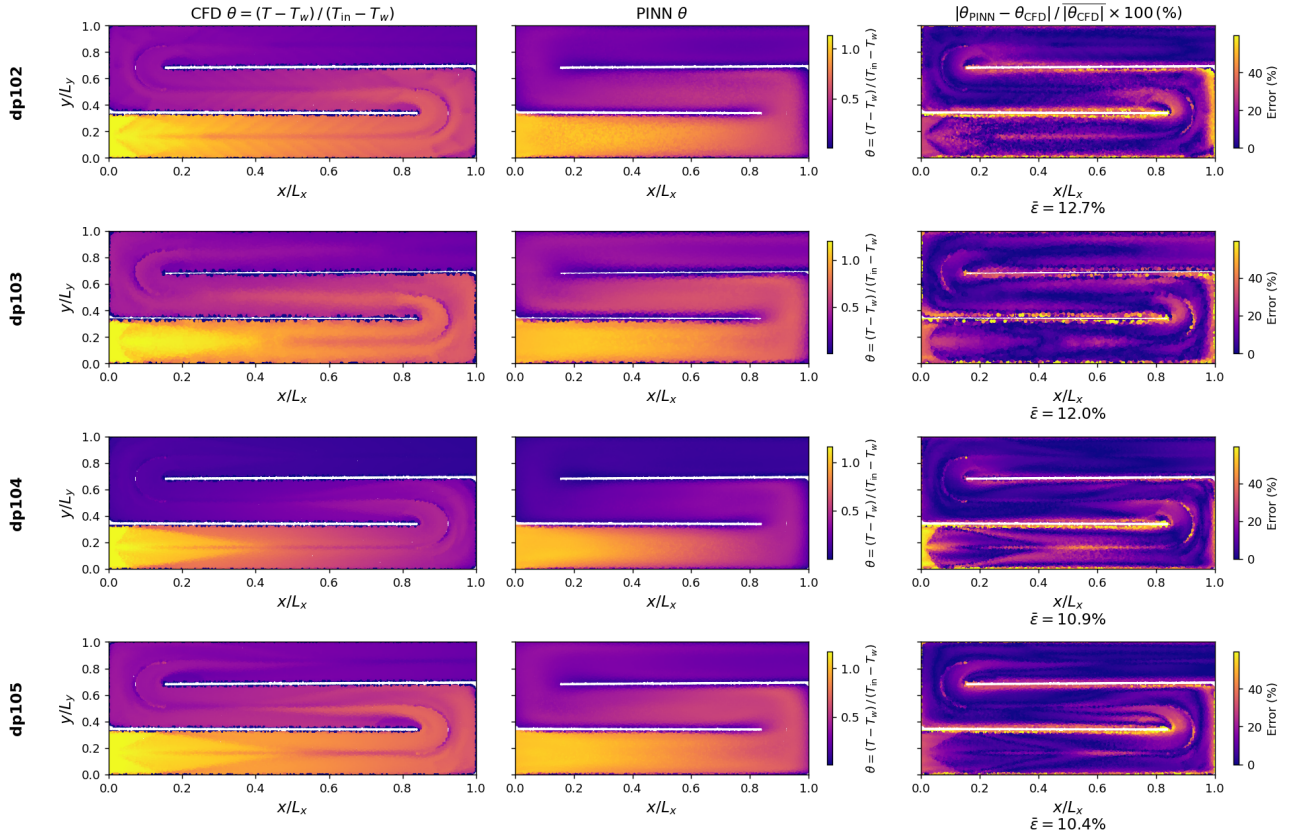


Figure 9: Normalised temperature θ : CFD (left), PINN V3 prediction (centre), relative error in % (right).

3.10.2 VELOCITY FIELD

The normalised streamwise velocity $v_{x,nd} = v_x/V_{in}$ shows mean errors of 27.2%, 22.2%, 26.6%, and 20.8%. The passage-to-passage structure is correctly reproduced: the model captures the flow acceleration in the narrower passages, the flow direction reversal between passages (positive v_x in Passage 1, negative in Passage 2, positive again in Passage 3), and the velocity deficit in the near-wall boundary layer. The highest errors appear immediately downstream of the ribs, where the rib wake induces

sharp recirculation zones with spatial scales well below the effective resolution of the network.

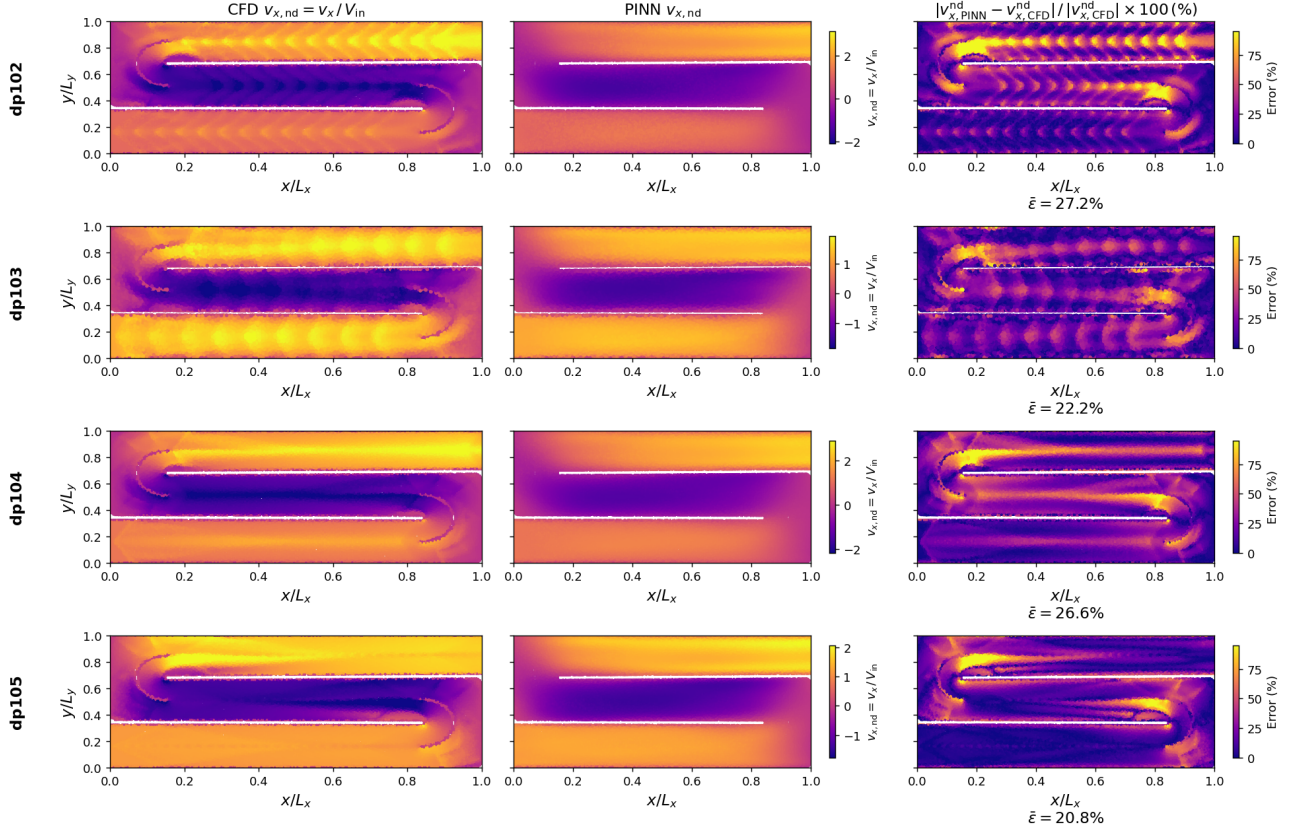


Figure 10: Normalised streamwise velocity $v_{x,nd}$: CFD (left), PINN V3 prediction (centre), relative error in % (right).

3.10.3 PRESSURE FIELD

The normalised pressure $\tilde{p} = (p - p_{\text{ref}})/\sigma_p$ is the best-predicted field overall, with mean errors of 8.4%, 6.7%, 5.6%, and 9.9%. The pressure decreases monotonically from inlet to outlet across the three passages, and the step change in pressure level between adjacent passages at the bends is correctly captured: a direct consequence of the sigmoid wall encoding introduced in PINN V3. Without it, this inter-passage pressure jump was entirely absent (see PINN V2 results, Fig. 7). The residual error pattern follows the rib spacing, reflecting the localised pressure drops induced by each rib that the smooth network partially smooths over.

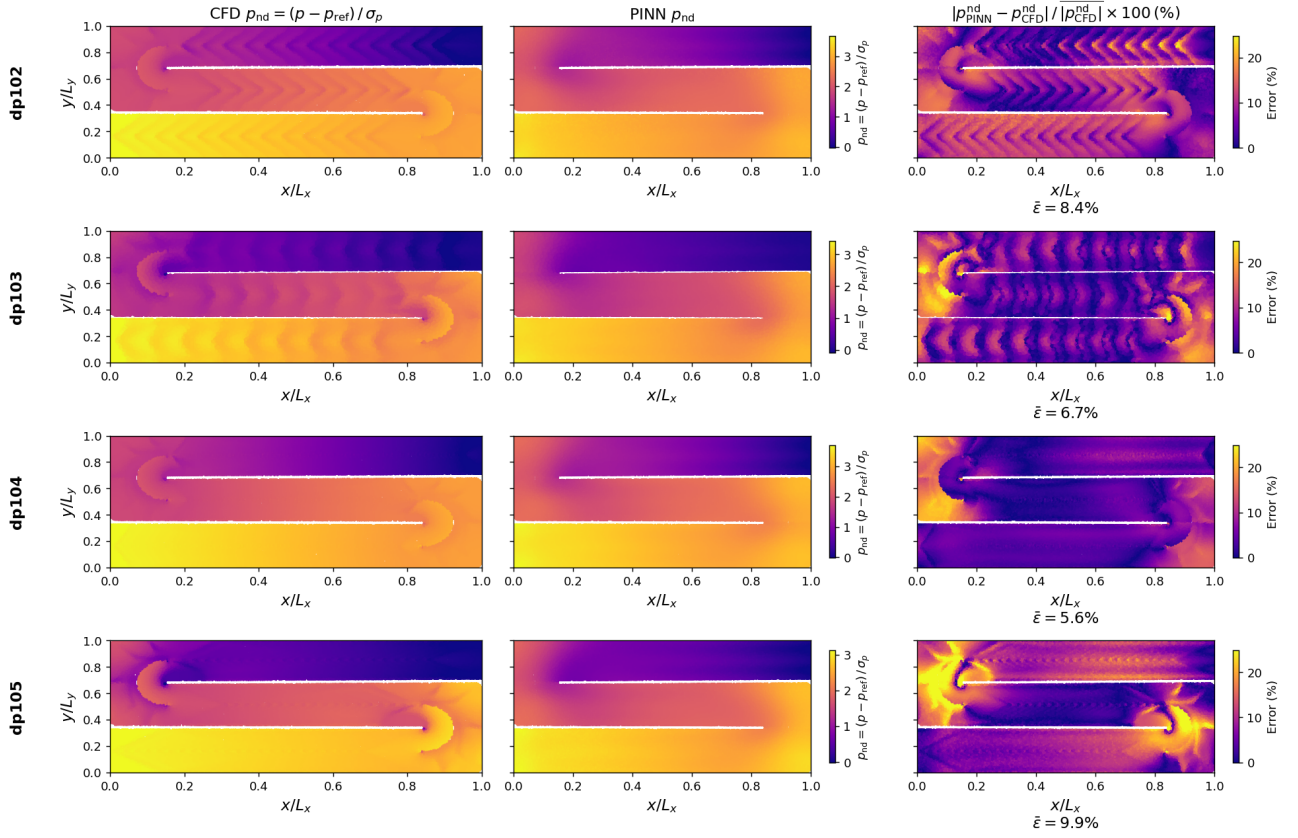


Figure 11: Normalised pressure $\tilde{p}$: CFD (left), PINN V3 prediction (centre), relative error in % (right).

4 MLP FOR WALL HEAT FLUX PREDICTION

4.1 MOTIVATION

The local wall heat flux is related to the temperature field by Fourier’s law:

$$q'' = -\kappa_f \left. \frac{\partial T}{\partial n} \right|_{\text{wall}} \quad (4.1)$$

where $\kappa_f = 2.61 \times 10^{-2}$ W/(m·K) is the fluid thermal conductivity and n is the wall-outward normal. Computing this from the PINN temperature field requires resolving the temperature gradient in the near-wall boundary layer, where T changes from the bulk value to T_w over the thermal boundary layer thickness $\delta_T \sim D_h/Nu$. For the Reynolds numbers considered here ($Re = 20,000$ – $200,000$), the Nusselt numbers $Nu \sim 100$ – 500 imply $\delta_T \sim 0.1$ – 0.3 mm, well below the spatial resolution achievable with a smooth PINN trained on 10,000 collocation points distributed across a ~ 0.6 m³ domain.

A fully data-driven Multilayer Perceptron (MLP) is therefore trained directly on CFD wall heat flux values. The training target is the normalised wall heat flux:

$$q^* = \frac{q''_w D_h}{\kappa_f |T_{\text{in}} - T_w|} \quad (4.2)$$

4.2 INPUT CONSTRUCTION

Wall heat flux data is loaded from the pre-processed CFD dataset. A buffer of 420 mm is trimmed at each end of the channel in x to ensure a fully-developed flow regime, removing the entrance and exit regions where the flow has not yet reached a periodic state. After trimming, each design point contributes on the order of 50,000–200,000 wall face centroids.

The wall heat flux is located on the bottom wall only ($z = 0$ is constant), so the z -coordinate carries no spatial information and is dropped. The MLP input for a wall point at normalised position $(x_{\text{nd}}, y_{\text{nd}})$ is:

$$\mathbf{x}_{\text{MLP}} = (x_{\text{nd}}, y_{\text{nd}}, AR_{\text{nd}}, (e/D_h)_{\text{nd}}, (P/e)_{\text{nd}}, \alpha_{\text{nd}}, Re_{\text{nd}}, s_1, s_2) \in \mathbb{R}^9 \quad (4.3)$$

where $x_{\text{nd}}, y_{\text{nd}}$ are z-score normalised coordinates, the five design parameters follow the same normalisation as in the PINN (Table 2), and s_1, s_2 are the sigmoid wall-encoding features from Eq. (3.13).

The heat flux output is also z-score normalised:

$$\tilde{q}'' = (q'' - \overline{q''}_{\text{train}}) / \sigma_{q'', \text{train}} \quad (4.4)$$

using the mean and standard deviation of raw heat flux over all training design points.

4.3 TRAINING PROCEDURE

Training uses a standard MSE loss over mini-batches of 8,192 points drawn by random permutation from all training design points. The Adam optimiser is used with L2 weight decay. The learning rate follows the same exponential decay schedule as the PINN. An early-stopping mechanism saves the best checkpoint when the test MSE improves by more than 10^{-5} and terminates training when no improvement is seen for 500 consecutive epochs (maximum 10,000 epochs). The train/test split is performed at the design point level: all wall points from a given design point are either entirely in training or entirely in test, ensuring the evaluation measures true generalisation to unseen geometries.

A hyperparameter search was conducted using Bayesian optimisation on the IZAR cluster (details in Appendix A). Training completes in approximately 7 minutes on GPU. Inference over all wall face centroids of a new design point takes under 1 second.

The selected hyperparameters are: hidden dimension 128, 8 layers ($\approx 330,000$ parameters), $lr_0 = 5.9 \times 10^{-4}$, $lr_{\text{end}} = 3.4 \times 10^{-4}$, weight decay 1.3×10^{-4} , batch size 8,192.

4.4 RESULTS ON UNSEEN DESIGN POINTS

Results are evaluated on the 4 design points held out from training (dp102, dp103, dp104, dp105). The mean relative error $\bar{\varepsilon}$ defined in Eq. (3.15) is used throughout, computed over all wall surface points of each test design point.

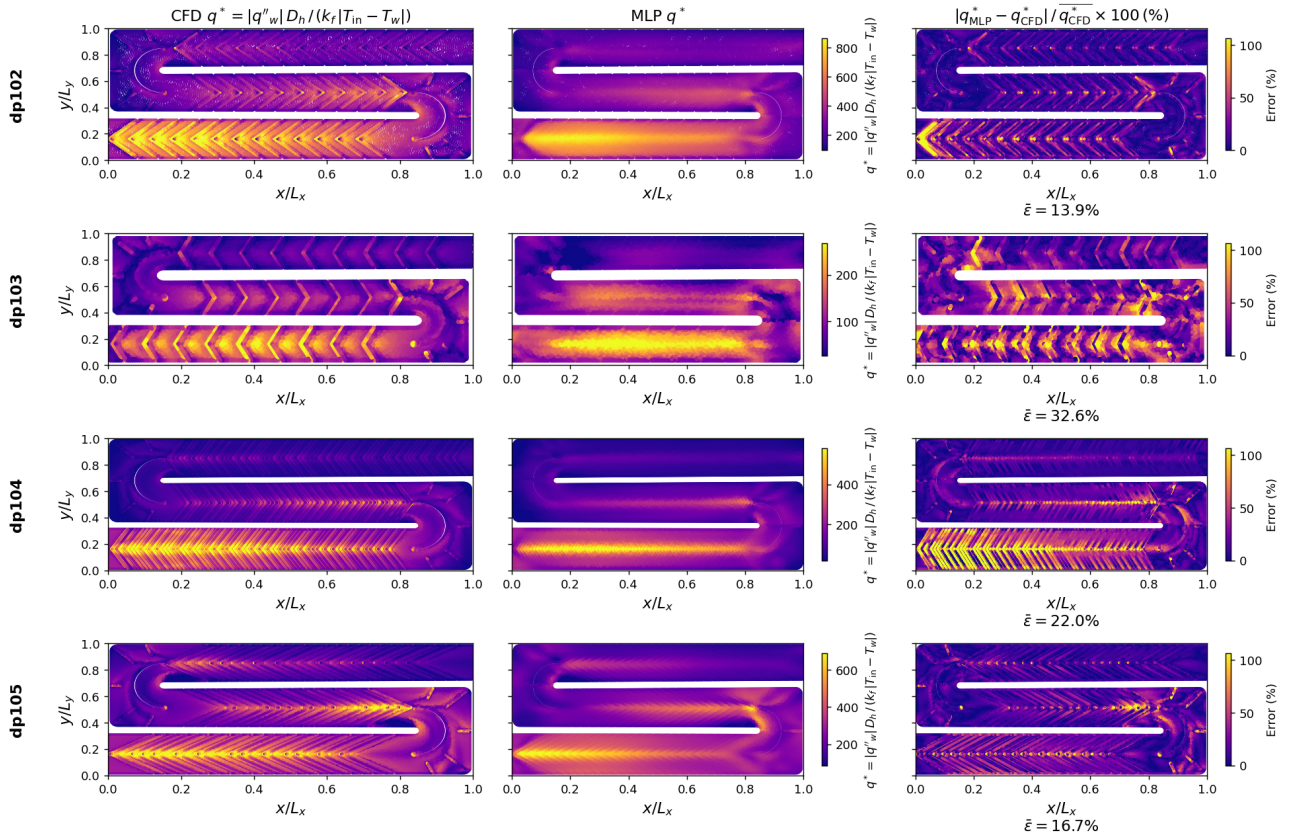


Figure 12: Normalised wall heat flux $q^* = q''_w D_h / (\kappa_f |T_{\text{in}} - T_w|)$: CFD (left), MLP prediction (centre), relative error in % (right) for 4 unseen design points. Mean relative errors: dp102: 13.9%, dp103: 32.6%, dp104: 22.0%, dp105: 16.7%.

Two observations stand out. First, the large-scale spatial distribution is correctly captured: the periodic rib pattern, the passage-to-passage variation, and the enhanced heat transfer near the bends all appear in the predicted field. Second, the MLP systematically smooths the sharp local heat flux peaks immediately at the rib edges, a consequence of the smooth function class (sine activations) that preferentially represents low spatial frequencies. This smoothing effect is consistent across all four test cases and becomes the bottleneck for accurate HTC prediction near the ribs. Finally, the error range (13.9%–32.6%) is wider than the PINN velocity error range (20.8%–27.2%), reflecting the absence of physical constraints: the MLP must interpolate in the design space using only the training data, and design points in less-represented regions of the parameter space (dp103: low- Re , low- AR corner) perform worse.

5 HEAT TRANSFER COEFFICIENT

5.1 COMBINED PREDICTION PIPELINE

The local Heat Transfer Coefficient is:

$$\text{HTC} = \frac{q_w''}{T_w - T_b} \quad (5.1)$$

where q_w'' is predicted by the MLP and T_b is the local bulk fluid temperature extracted from the PINN volume temperature field.

5.2 BULK TEMPERATURE COMPUTATION

T_b is computed by averaging the PINN temperature field over six key cross-sectional planes and then interpolating:

- T_0 : inlet ($x = 0$ m, Passage 1)
- T_1 : end of Passage 1 ($x = 0.9395$ m)
- T_2 : start of Passage 2 ($x = 0.9395$ m)
- T_3 : end of Passage 2 ($x = 0.1870$ m)
- T_4 : start of Passage 3 ($x = 0.1870$ m)
- T_5 : outlet ($x = 1.1320$ m)

Each plane temperature is the mean over all PINN prediction points within ± 6 mm of the plane position and within the appropriate y -band for that passage. T_b at each wall point is then computed piecewise:

$$T_b = T_0 + \frac{T_1 - T_0}{X_{\text{top}}} \cdot x (\text{Passage 1}) \quad (5.2)$$

$$T_b = T_2 + \frac{T_3 - T_2}{X_{\text{top}} - X_{\text{bot}}} \cdot (X_{\text{top}} - x) (\text{Passage 2}) \quad (5.3)$$

$$T_b = T_4 + \frac{T_5 - T_4}{L_{P3}} \cdot (x - X_{\text{bot}}) (\text{Passage 3}) \quad (5.4)$$

In the 180° bend regions, an arc-based interpolation is used. For the top bend ($x \geq X_{\text{top}}$, $y \leq y_{P23}$):

$$T_b = \frac{T_1 + T_2}{2} + \frac{T_2 - T_1}{\pi} \arcsin\left(\frac{y - c_{y,\text{top}}}{r}\right) \quad (5.5)$$

where $r = \sqrt{(x - X_{\text{top}})^2 + (y - c_{y,\text{top}})^2}$ is the arc radius. The bottom bend is handled analogously. This formulation correctly accounts for the change in flow direction across the bend, mapping the streamwise position along the arc to a linear interpolation between the upstream and downstream plane temperatures.

The Nusselt number for each section is then:

$$Nu = \frac{\text{HTC} \cdot D_h}{\kappa_f} \quad (5.6)$$

using the section-average hydraulic diameter D_h from the CFD geometry registry.

5.3 RESULTS

Results are reported using the mean relative error $\bar{\varepsilon}$ of Eq. (3.15). Table 3 presents section-averaged Nusselt numbers, heat fluxes, and bulk temperatures across all four test design points and all five channel sections.

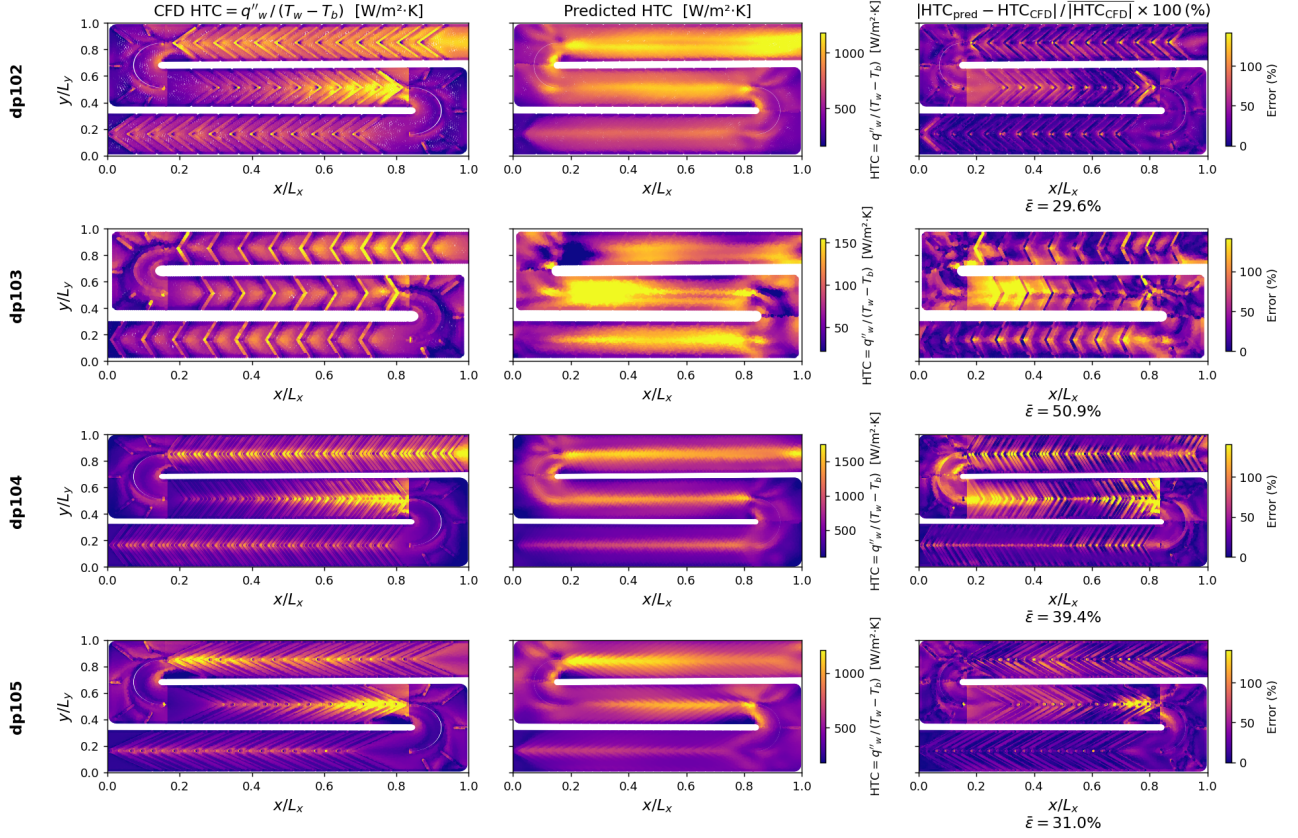


Figure 13: Heat Transfer Coefficient $[W/m^2 \cdot K]$: CFD reference (left), PINN+MLP prediction (centre), relative error in % (right) for 4 unseen design points. Mean relative errors: dp102: 29.6%, dp103: 50.9%, dp104: 39.4%, dp105: 31.0%. Local errors up to 100% occur at rib edges; large-scale variation and inter-design-point ordering are correctly captured.

Several patterns emerge from Table 3. The Nusselt number is systematically over-predicted in all sections and all design points, with errors ranging from 20% in well-resolved straight passages to over 70% in Passage 2 of dp103. This over-prediction reflects the compound effect of the MLP smoothing the heat flux peaks (raising q''_{pred} relative to sharp-rib CFD values in the average) and the PINN over-estimating the bulk temperature gradient (raising T_b relative to CFD, which reduces $T_w - T_b$ and inflates HTC). The bulk temperature planes are predicted with good accuracy ($< 1.4\%$ error in all cases), confirming that the PINN's section-averaged temperature is reliable even when local pointwise errors are higher. The bends consistently show higher Nusselt errors than the straight passages, reflecting the higher difficulty of resolving the three-dimensional turning flow.

Table 3: Section-averaged Nusselt numbers, wall heat fluxes, and bulk temperatures predicted by the PINN+MLP pipeline vs. CFD on 4 unseen design points. Nu_{norm} is the Nusselt number normalised by its smooth-channel reference. ε denotes the relative error (%).

DP	Section	Nu_{norm}		ε_{Nu}	q'' [W/m ²]		$\varepsilon_{q''}$	T_{in} [K]		$\varepsilon_{T_{\text{in}}}$	T_{out} [K]		$\varepsilon_{T_{\text{out}}}$
		CFD	Pred	(%)	CFD	Pred	(%)	CFD	Pred	(%)	CFD	Pred	(%)
dp102	Pass 1	1.856	2.076	19.6	13202	14501	9.8	330.0	330.0	0.00	315.8	311.8	1.27
	Pass 2	2.285	2.865	38.1	7453	9062	21.6	311.6	307.4	1.36	306.2	303.2	0.96
	Pass 3	2.743	3.287	22.9	5146	6189	20.3	303.9	301.3	0.85	300.3	299.5	0.25
	Top Bend	1.427	1.998	41.3	6086	8482	39.4	315.8	311.8	1.27	311.6	307.4	1.36
	Bot. Bend	1.630	2.366	45.9	3851	5596	45.3	306.2	303.2	0.96	303.9	301.3	0.85
dp103	Pass 1	2.440	3.410	45.2	1861	2556	37.4	329.8	329.4	0.12	317.5	313.8	1.14
	Pass 2	2.703	4.360	70.9	1169	1830	56.6	313.2	311.1	0.66	307.8	305.9	0.63
	Pass 3	2.861	2.958	37.2	748	789	5.4	305.3	303.6	0.56	301.3	301.6	0.09
	Top Bend	2.072	3.191	63.3	1106	1707	54.4	317.5	313.8	1.14	313.2	311.1	0.66
	Bot. Bend	2.364	3.074	43.5	765	1004	31.2	307.8	305.9	0.63	305.3	303.6	0.56
dp104	Pass 1	1.891	2.210	28.6	9946	11260	13.2	327.6	327.3	0.12	307.7	305.7	0.64
	Pass 2	2.698	3.281	53.0	4809	5365	11.6	304.1	302.4	0.56	299.0	298.1	0.31
	Pass 3	3.267	3.974	32.0	2053	2539	23.7	297.8	296.3	0.49	295.5	295.5	0.01
	Top Bend	1.440	1.990	41.5	3635	5023	38.2	307.7	305.7	0.64	304.1	302.4	0.56
	Bot. Bend	1.748	2.712	58.4	1643	2529	54.0	299.0	298.1	0.31	297.8	296.3	0.49
dp105	Pass 1	1.554	1.865	22.6	10429	12479	19.7	329.5	329.4	0.01	316.8	313.4	1.06
	Pass 2	2.156	2.655	39.7	7297	8739	19.8	311.7	309.2	0.78	306.8	304.5	0.76
	Pass 3	2.456	3.153	29.5	4695	6034	28.5	304.1	302.6	0.50	300.8	299.7	0.35
	Top Bend	1.580	2.059	33.5	6835	8864	29.7	316.8	313.4	1.06	311.7	309.2	0.78
	Bot. Bend	1.819	2.329	31.0	4486	5740	28.0	306.8	304.5	0.76	304.1	302.6	0.50

6 DISCUSSION AND CONCLUSION

6.1 WHAT WAS DONE

This project built an end-to-end surrogate pipeline for the prediction of flow fields and heat transfer in rib-turbulated gas turbine blade cooling channels, covering the full modelling chain from raw CFD data to a deployable HTC map over an unseen design point.

Loss normalisation and the weighting problem. A central contribution is the demonstration that the PINN loss weighting problem, which requires balancing PDE residuals, boundary conditions, and supervised data terms that can differ by several orders of magnitude across design points and field variables, is solved automatically by proper non-dimensionalisation. Normalising each supervised and BC loss term by the variance of the corresponding field from CFD training data, and scaling PDE residuals by coordinate and output standard deviations, produces $\mathcal{O}(1)$ values for all loss components with uniform weights. Three adaptive weighting strategies (manual tuning, learnable Lagrangian weights, SoftAdapt) were tested and found to provide no meaningful improvement at this stage.

Adaptive collocation point sampling. Uniform volume sampling failed to enforce the wall boundary conditions adequately. A three-pool strategy was implemented: 50% of collocation points drawn uniformly from the interior, 15% placed within 1 mm of horizontal wall faces, and 35% placed within 20 mm of vertical wall faces. The thresholds were chosen proportionally to the channel dimensions, keeping the near-wall fractions comparable in relative terms across both directions.

Sigmoid wall-position encoding. The most impactful single contribution to PINN V3 is the introduction of two sigmoid features s_1, s_2 encoding passage membership. Without these features (PINN V2), the pressure field completely failed on the multi-pass geometry, producing a smooth field that ignored the internal walls. With them (PINN V3), the pressure is correctly structured across all three passages, with mean errors of 5.6%–9.9% on unseen design points. The encoding is lightweight (two scalar inputs, no additional network parameters) and transfers across all 50 design points since the wall positions are fixed.

Parametric training on 50 design points. A single PINN conditioned on five design parameters is trained simultaneously on all 50 CFD cases, enabling genuine generalisation to unseen geometries. Results on 4 held-out design points show: temperature field $\bar{\varepsilon} \approx 10$ –13%, streamwise velocity $\bar{\varepsilon} \approx 20$ –27%, and pressure drop $\bar{\varepsilon} \approx 6$ –10%.

Data-driven heat flux MLP. Because the PINN’s near-wall resolution is insufficient for Fourier’s law to be applied reliably, a separate MLP surrogate is trained directly on CFD wall heat flux data. The MLP uses the same design parameter conditioning and wall sigmoid encoding as the PINN, plus the two spatial coordinates of the wall surface. It predicts the normalised heat flux with $\bar{\varepsilon} = 14$ –33% on unseen design points, in under 1 second per inference.

Combined HTC surrogate. The PINN provides the bulk temperature T_b (via cross-sectional plane averaging and piecewise linear/arc interpolation), and the MLP provides the wall heat flux q''_w . Together they form a complete HTC surrogate. Section-averaged results (Table 3) show that bulk temperatures are very well predicted ($< 1.4\%$ error in all sections), while Nusselt numbers are systematically over-predicted by 20%–70%, with the bends showing the largest errors. Despite absolute inaccuracies, the inter-design-point ordering of HTC is correctly preserved, which is the minimum requirement for integration into an optimisation loop.

6.2 LIMITATIONS

Both models struggle to resolve the sharp spatial features induced by the ribs. Rib wakes, heat flux peaks at rib edges, and the thin thermal boundary layer all vary on sub-millimetre scales that smooth

sine-activated networks cannot capture accurately. This spectral bias is the primary source of error in both the PINN velocity field (20–27%) and the MLP heat flux (14–33%), and it compounds in the HTC to produce errors of 30–70%.

Additionally, the MLP’s wider error range across design points compared to the PINN reflects the absence of physical constraints: design points in less-sampled regions of the five-dimensional parameter space (in particular low- Re , low- AR) are predicted less reliably.

Finally, the PINN could not be run in a fully data-free setting. As shown in the PINN V1 experiments, removing the sparse supervised CFD data causes the pressure field to collapse to a trivially constant solution, which satisfies both the momentum equations and the outlet boundary condition simultaneously. Velocity and temperature remain reasonably well-constrained by the physics alone, but pressure requires at least a small number of labelled points to select the physically correct solution. This dependence on sparse CFD supervision means the surrogate cannot be queried for a completely new design point without first running at least a partial CFD simulation, which limits its applicability as a true zero-data surrogate.

6.3 FUTURE WORK

Several directions are identified for improving the surrogate models:

Domain decomposition (XPINN) [11]. Splitting the domain into dedicated sub-networks for the boundary layer, the straight passage interior, and the bend regions would allow each sub-network to be trained at the spatial resolution appropriate to its local physics. Heat flux could then be derived directly from the near-wall sub-network’s temperature gradient via Fourier’s law, eliminating the need for a separate MLP and ensuring physical consistency. Interface continuity conditions connect the sub-networks and must be enforced carefully.

Bend encoding. The sigmoid wall encoding proved effective at disambiguating the three passages. A natural extension is to apply a similar approach to the bend regions, for instance using a curve-based distance feature that encodes the proximity of each point to the 180° bend arc, giving the network explicit geometric awareness of the regions where flow direction reverses and errors are currently largest.

Reducing the dependence on supervised data. In PINN V1, pressure could not be recovered without supervised data, even in a simple straight channel. An alternative strategy would be to use the PINN-predicted velocity and temperature fields (which are already well-constrained by the NS and energy equations) as inputs to a secondary network or post-processing step to infer the pressure field, rather than relying on direct sparse pressure supervision. This could reduce the amount of CFD data required per design point.

Recurrent models. The current PINN treats the three-passage channel as a single monolithic domain. A recurrent or sequential formulation, where the solution in each passage is conditioned on the exit conditions of the previous passage, could provide a more structured inductive bias for the multi-pass geometry.

More advanced turbulence modelling. The current model uses steady-state incompressible NS equations, which implicitly relies on the CFD’s turbulence modelling being absorbed into the supervised data. An alternative would be to include Reynolds-stress terms or the k - ε transport equations explicitly in the PDE loss, which could improve accuracy at high Reynolds numbers where turbulent mixing becomes the dominant heat transfer mechanism.

Integration into optimisation. The immediate practical next step is embedding the surrogate pipeline into the Bayesian optimisation framework developed at LAMD, performing a full multi-objective search over the five-parameter design space with pressure drop and HTC as competing objectives, to produce a Pareto frontier of optimal rib configurations.

APPENDIX A HYPERPARAMETER SEARCH DETAILS

A.1 PINN HYPERPARAMETER SWEEP

The PINN hyperparameter search used Bayesian optimisation via Weights & Biases on the EPFL IZAR HPC cluster. The metric minimised was the total training loss `loss/total`. The following hyperparameters were varied:

- `hidden_dim`: {64, 128, 256}
- `n_layers`: {4, 6, 8}
- `n_train` (collocation points per epoch per DP): {5,000, 10,000, 20,000}
- `epochs`: {500, 1,000, 2,000, 5,000}
- `lr`: log-uniform in $[10^{-3}, 10^{-2}]$
- `lr_decay_ratio`: log-uniform in $[0.01, 0.5]$

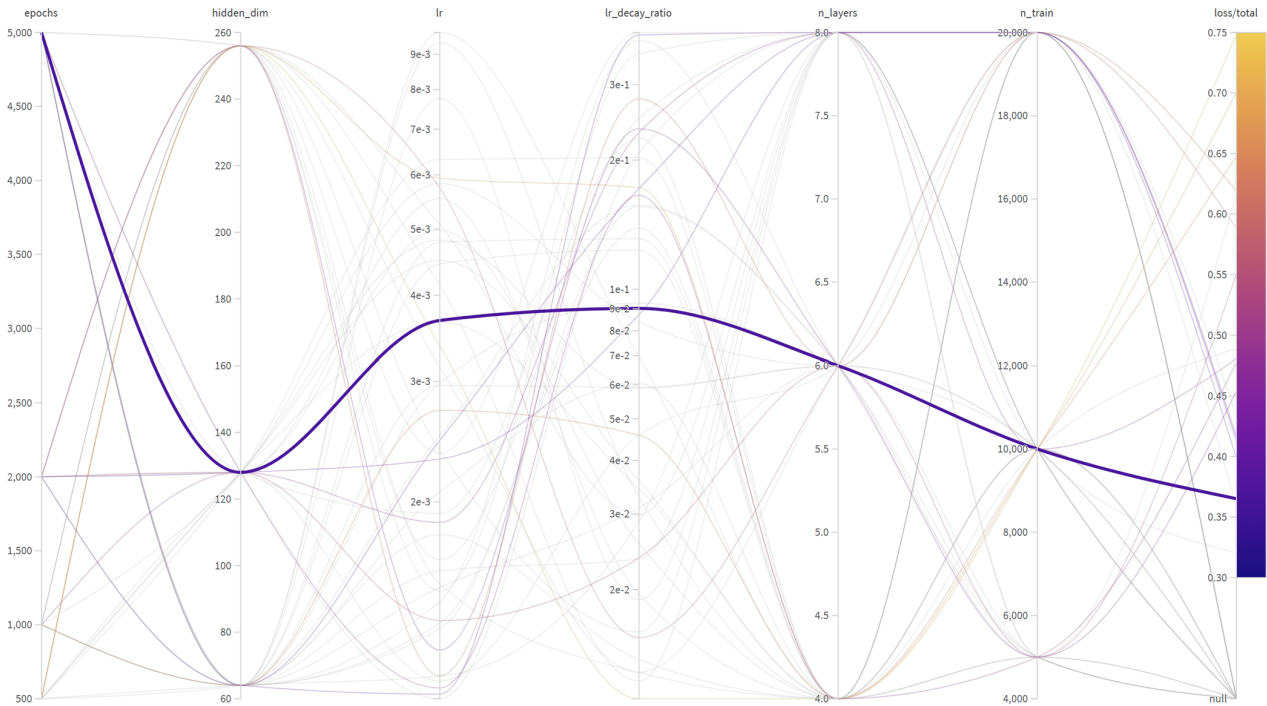


Figure 14: PINN hyperparameter sweep parallel coordinates plot. Each line is one trial coloured by total training loss (dark blue = low/good, yellow = high/poor). The bold dark line shows the selected configuration. Epochs and number of layers are the most discriminating hyperparameters.

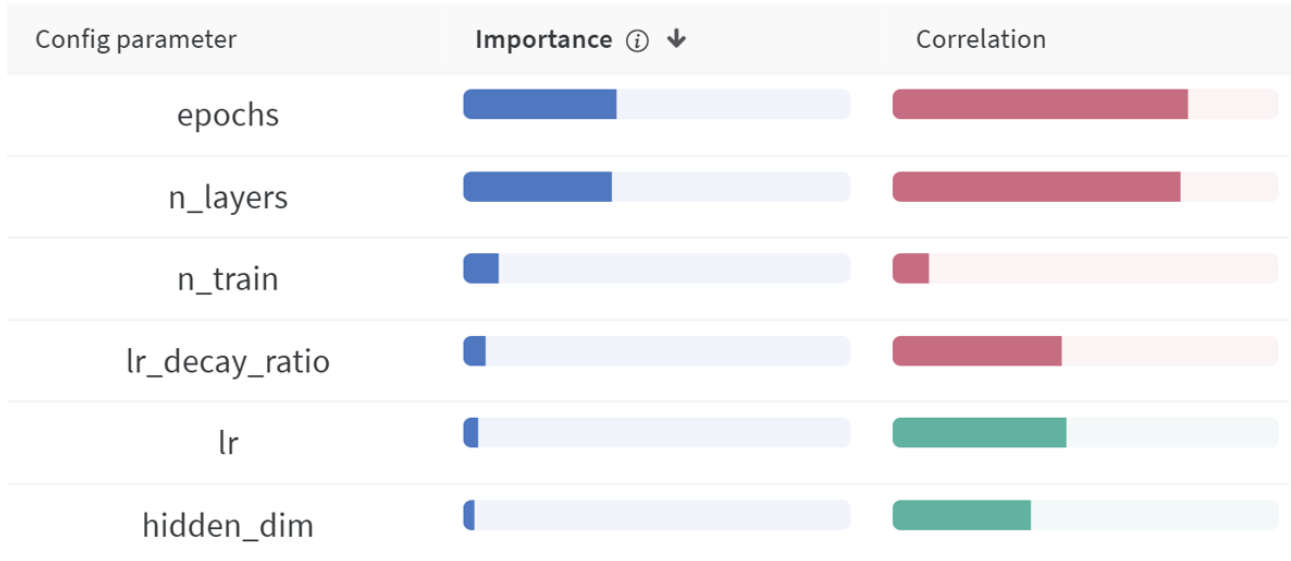


Figure 15: PINN hyperparameter importance ranking from the Bayesian sweep. Epochs and number of layers are the most discriminating hyperparameters; low-loss trials consistently use the maximum 5,000 epochs and 6+ layers. The parameter `n_train` shows lower importance, suggesting the model is not strongly collocation-point-limited in the 5,000–20,000 range.

The importance analysis confirms that `epochs` and `n_layers` are the dominant hyperparameters: low-loss trials consistently use the maximum 5,000 epochs and 6+ layers. The parameter `n_train` shows lower importance, suggesting the model is not strongly collocation-point-limited in the 5,000–20,000 range.

A.2 MLP HYPERPARAMETER SWEEP

The MLP hyperparameter search also used Bayesian optimisation. The metric minimised was the best test normalised MSE `best_test_mse_nd`. The following hyperparameters were varied:

- `hidden_dim`: {8, 64, 128}
- `n_layers`: {4, 8}
- `lr`: log-uniform in $[3 \times 10^{-4}, 5 \times 10^{-3}]$
- `lr_end`: log-uniform in $[10^{-5}, 5 \times 10^{-4}]$
- `weight_decay`: log-uniform in $[10^{-6}, 10^{-2}]$

Maximum 1,000 epochs with patience 300 per trial.

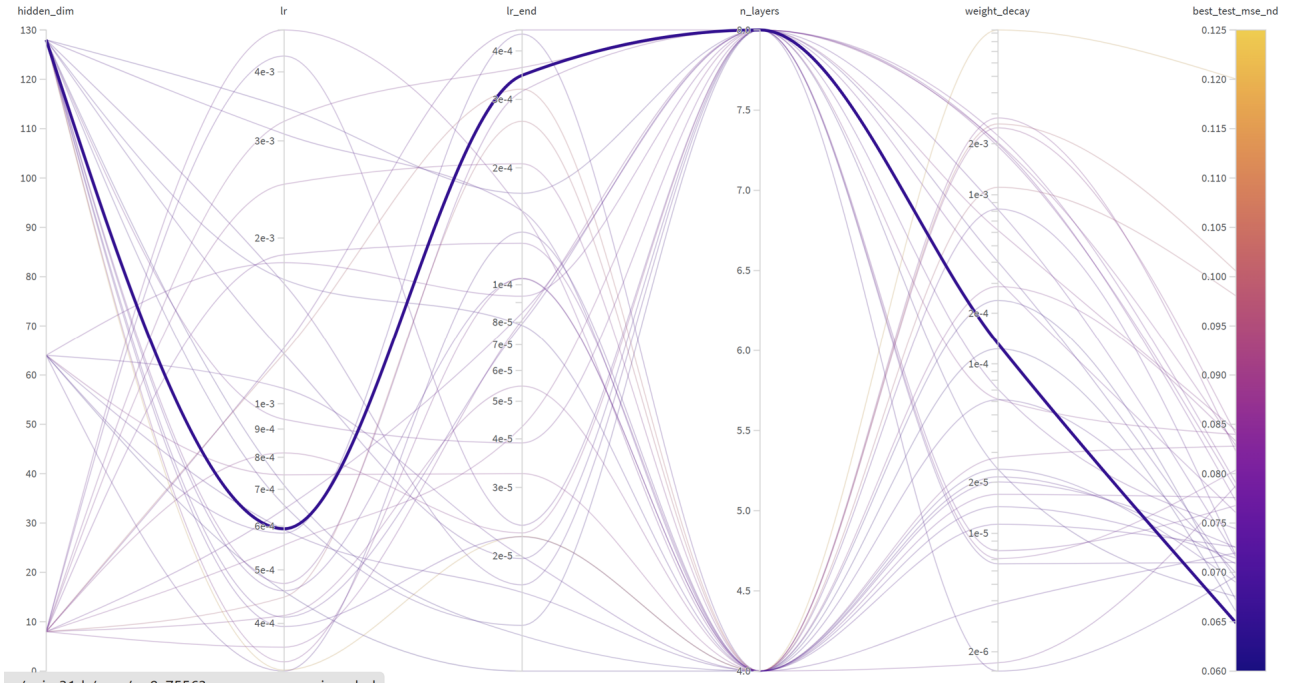


Figure 16: MLP hyperparameter sweep parallel coordinates plot. **weight_decay** and **hidden_dim** dominate the importance ranking, confirming that L2 regularisation is the primary mechanism preventing overfitting when training on a large combined dataset.

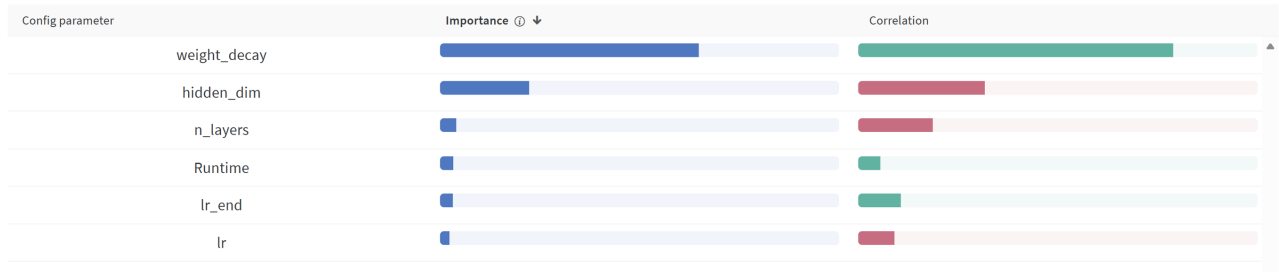


Figure 17: MLP hyperparameter importance ranking from the Bayesian sweep. **weight_decay** and **hidden_dim** are the most discriminating hyperparameters, confirming that L2 regularisation is the primary mechanism preventing overfitting when training on a large combined dataset.

APPENDIX B NON-DIMENSIONALISATION OF THE NS EQUATIONS

Starting from the dimensional steady incompressible NS equation:

$$\mathbf{v} \cdot \nabla v_i = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \nu \tilde{\mu} \nabla^2 v_i \quad (\text{B.1})$$

The velocity and pressure are non-dimensionalised as:

$$\tilde{v}_i = \frac{v_i}{V_{\text{in}}}, \quad \tilde{p} = \frac{p - p_{\text{ref}}}{\sigma_p}, \quad \sigma_p = \text{std}(p_{\text{CFD}}) \quad (\text{B.2})$$

Substituting and dividing through by V_{in}^2 gives:

$$\tilde{\mathbf{u}} \cdot \nabla \tilde{u}_i + \frac{\sigma_p}{\rho V_{\text{in}}^2} \partial_{x_i} \tilde{p} - \frac{\nu \tilde{\mu}}{V_{\text{in}}} \nabla^2 \tilde{u}_i = 0 \quad (\text{B.3})$$

The residual actually minimised in the PINN loss sets the coefficient $\sigma_p/(\rho V_{\text{in}}^2)$ to unity:

$$\tilde{\mathbf{u}} \cdot \nabla \tilde{u}_i + \partial_{x_i} \tilde{p} - \frac{\nu \tilde{\mu}}{V_{\text{in}}} \nabla^2 \tilde{u}_i = 0 \quad (\text{B.4})$$

This is not a physical approximation. The two forms share exactly the same zero set: any $(\tilde{\mathbf{u}}, \tilde{p})$ that makes one residual zero also makes the other zero, since $\sigma_p/(\rho V_{\text{in}}^2) \neq 0$. The difference only manifests during training, where the residual is non-zero. Setting the coefficient to unity keeps the pressure gradient term $\mathcal{O}(1)$ and equally weighted relative to the advection and diffusion terms regardless of the flow regime, which is the intended behaviour. Using the formally correct coefficient $\sigma_p/(\rho V_{\text{in}}^2)$ would instead amplify the pressure gradient contribution for high-friction design points where $\sigma_p \gg \rho V_{\text{in}}^2$, biasing the optimiser toward satisfying the pressure balance at the expense of the momentum balance.

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